



Graphs in Machine Learning

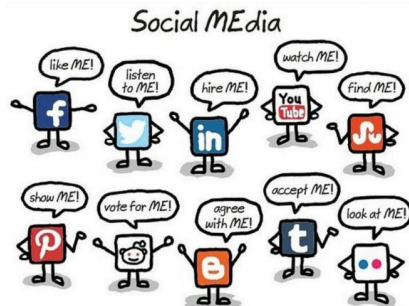
Michal Valko

INRIA Lille - Nord Europe, France

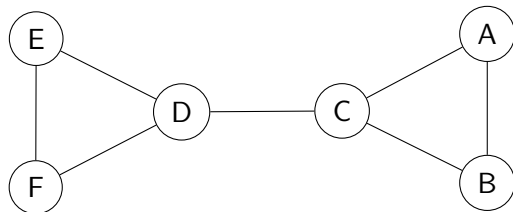
Partially based on material by: Andreas Krause, Branislav Kveton, Michael Kearns

Graphs from **social networks**

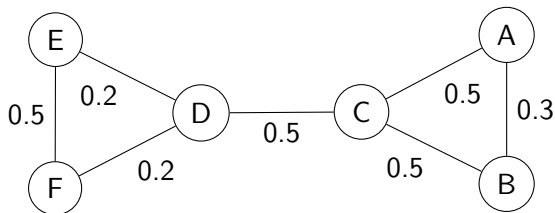
- ▶ people and their interactions
- ▶ directed (Twitter) and undirected (Facebook)
- ▶ structure is rather a *phenomena*
- ▶ typical ML tasks
 - ▶ advertising
 - ▶ product placement
 - ▶ link prediction (PYMK)



Success story #1 **Product placement** - problem



Success story #1 Product placement - problem



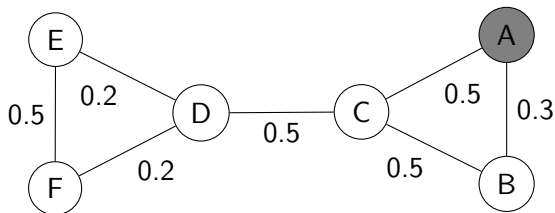
Who should get free cell phones?

$V = \{\mathbf{A}lice, \mathbf{B}ob, \mathbf{C}harlie, \mathbf{D}orothy, \mathbf{E}ric, \mathbf{F}iona\}$

$F(A)$ = Expected number of people influenced when targeting A
(under some propagation model)

How would you choose the target customers?

Success story #1 Product placement - problem

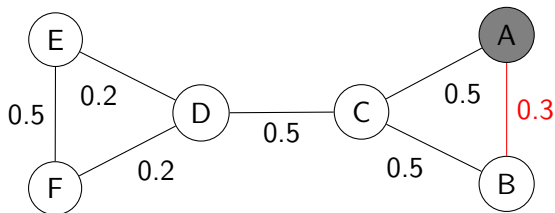


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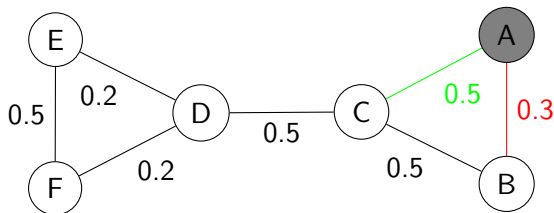


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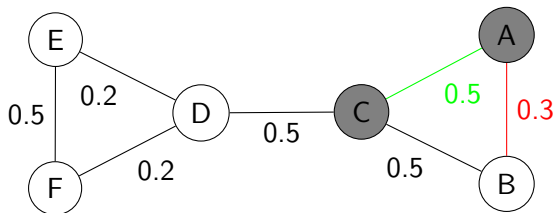


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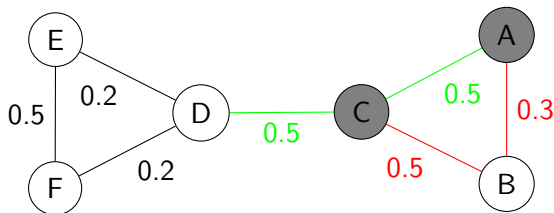


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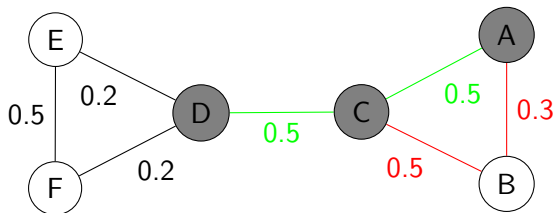


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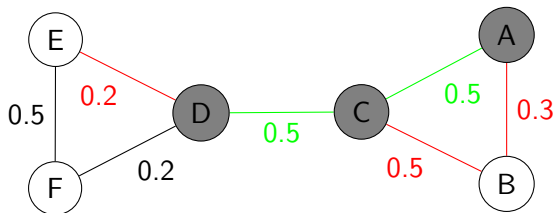


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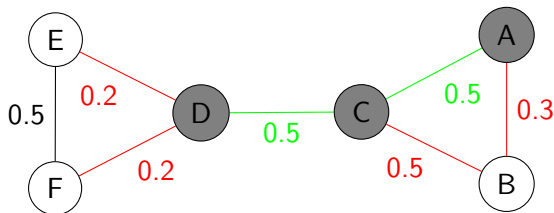


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Submodularity

A set function on a discrete set A is **submodular** if for any $S \subseteq T \subseteq A$ and for any $e \in A$

$$f(S \cup \{e\}) - f(S) \geq f(T \cup \{e\}) - f(T)$$

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Special case: f is also **nonnegative** and **monotone**.

Objective: Find $\arg \max_{S \subseteq A, |S| \leq k} f(S)$

Property: NP-hard in general

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Obj

Prop

How does this relate to our problem
of **product placement** on a **social
network graph**?

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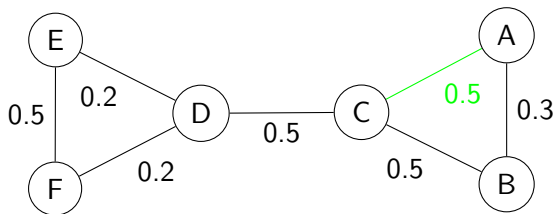
Objective: Find $\arg \max_{S \subseteq A, |S| \leq k} f(S)$

Property: NP-hard in general

Let $S^* = \arg \max_{S \subseteq A, |S| \leq k} f(S)$ and let S be the **greedy solution**.

$$\text{Then } f(S) \leq \left(1 - \frac{1}{e}\right) \cdot f(S^*).$$

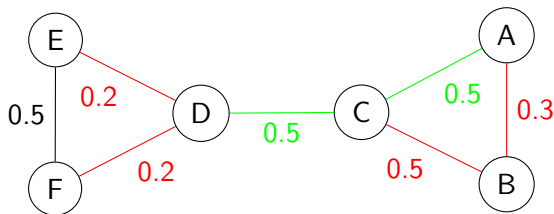
Success story #1 Product placement - solution



Key idea: Flip coins c in advance $\rightarrow$ “live” edges

(cf. Andreas Krause <http://submodularity.org/>)

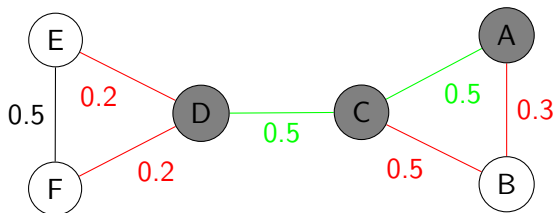
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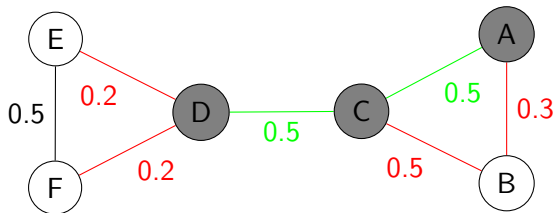
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 $F_c(A)$ = People influenced under outcome c (set cover!)

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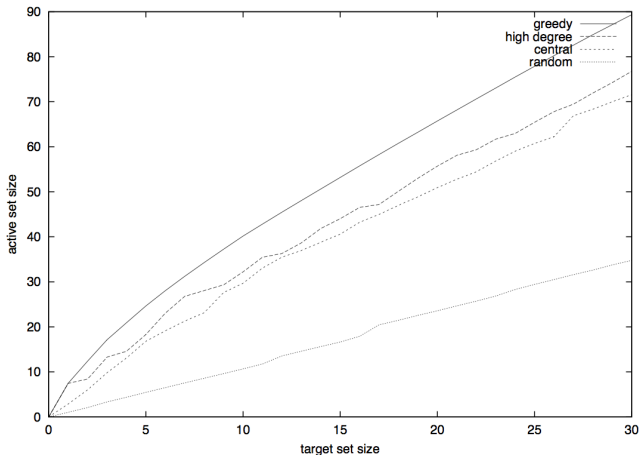
$F_c(A)$ = People influenced under outcome c (set cover!)

$F(A) = \sum_c P(c)F_c(A)$ is submodular as well!

(cf. Andreas Krause <http://submodularity.org/>)

Success story #1 **Product placement** - comparison

propagation on the ArXiv/Physics co-authorship dataset



Graphs from **utility** and **technology** networks

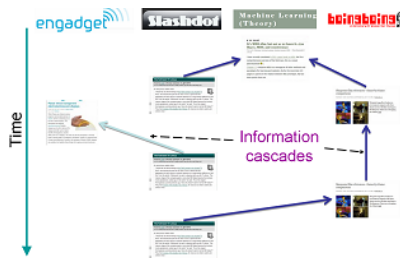
- ▶ link services
- ▶ power grids, roads, Internet, sensor networks
- ▶ structure is either *hand designed* or not
- ▶ typical ML tasks
 - ▶ best routing under unknown or variable costs
 - ▶ identify the node of interest



Berkley's Floating Sensor Network

Graphs from **information networks**

- ▶ web
- ▶ blogs
- ▶ wikipedia
- ▶ typical ML tasks
 - ▶ find influential sources
 - ▶ search (pagerank)



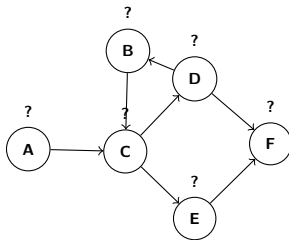
Blogs cascades (ETH) -
submodularity

Success story #2 Google PageRank

Objective: **Rank** all web pages (nodes on the graph) by how **many** other pages link to them and how **important** they are.

basic PageRank is independent of query and the page content

Internet $\rightarrow$ graph $\rightarrow$ matrix $\rightarrow$ stochastic matrix M



M is **stochastic**. We look for $Mv = v$, a steady-state vector.

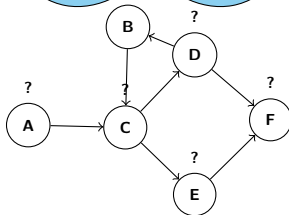
Success story #2 Google Rank

Objective: Rank (a web page) by how many other pages link to it.

basic PageRank

Internet $\rightarrow$ graph

Random Surfer Process



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Success story #2 Google PageRank

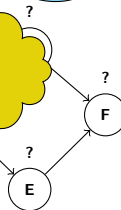
Objective: **Rank** (pages) by how **many** other pages link to it.

basic PageRank algorithm: follow random links to find interesting content

Internet $\rightarrow$ graph

Random Surfer Process

What is wrong with it?



M is **stochastic**. We look for $Mv = v$, a steady-state vector.

Success story #2 Google PageRank

<http://infolab.stanford.edu/~backrub/google.html>:

PageRank can be thought of as a model of user behavior. We assume there is a "random surfer" who is given a web page at random and keeps clicking on links, never hitting "back" but eventually gets bored and starts on another random page.

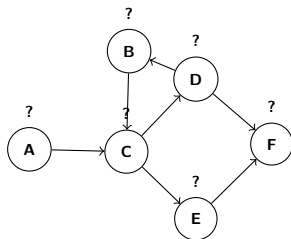
Google matrix: $G = (1 - p)M + p \cdot \frac{1}{n} \mathbb{1}_{n \times n}$, where $p = 0.15$

Success story #2 Google PageRank

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G is **still stochastic** (why?). We look for $Gv = 1 \times v$, steady-state vector, a right eigenvector with eigenvalue 1.

Perron's theorem: Such v exists and it is **unique** (if the entries of G are positive).

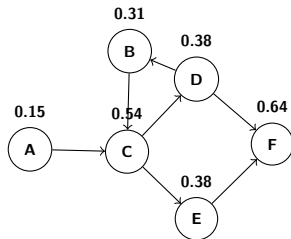


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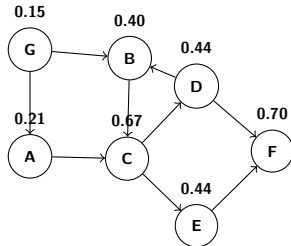


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Success story #2 Google PageRank

Problem: Find left eigenvector of a stochastic matrix.

History: [Desikan, 2006]

- ▶ The anatomy of a large-scale hypertextual web search engine [Brin & Page 1998]
- ▶ US patent for PageRank granted in 2001
- ▶ Google indexes 10's of billions of web pages ($1 \text{ billion} = 10^9$)
- ▶ Google serves ≥ 200 million queries per day
- ▶ Each query processed by ≥ 1000 machines
- ▶ All search engines combined process more than 500 million queries per day

Success story #2 Google PageRank

Problem: Find an eigenvector of a stochastic matrix.

- ▶ $n = 10^9$!!!
- ▶ luckily: **sparse** (average outdegree: 7)
- ▶ better than a simple centrality measure (e.g. degree)
- ▶ power method

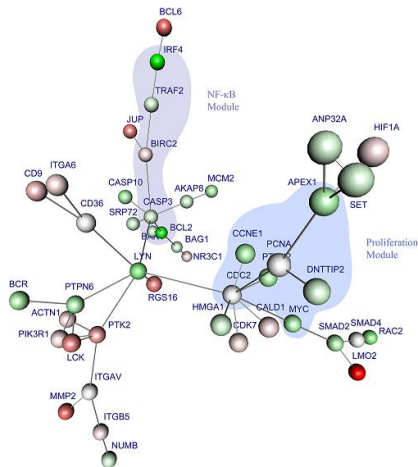
$$\mathbf{v}_0 = (1_A \quad 0_B \quad 0_C \quad 0_D \quad 0_E \quad 0_F)^\top$$

$$\mathbf{v}_1 = \mathbf{G}\mathbf{v}_0$$

$$\mathbf{v}_{t+1} = \mathbf{G}\mathbf{v}_t = \mathbf{G}^{t+1}\mathbf{v}$$

Graphs from **biological networks**

- ▶ protein-protein interactions
- ▶ gene regulatory networks
- ▶ typical ML tasks
 - ▶ discover unexplored interactions
 - ▶ learn or reconstruct the structure



Diffuse large B-cell lymphomas -
Dittrich et al. (2008)

Graphs from **similarity networks**

graph is not naturally given

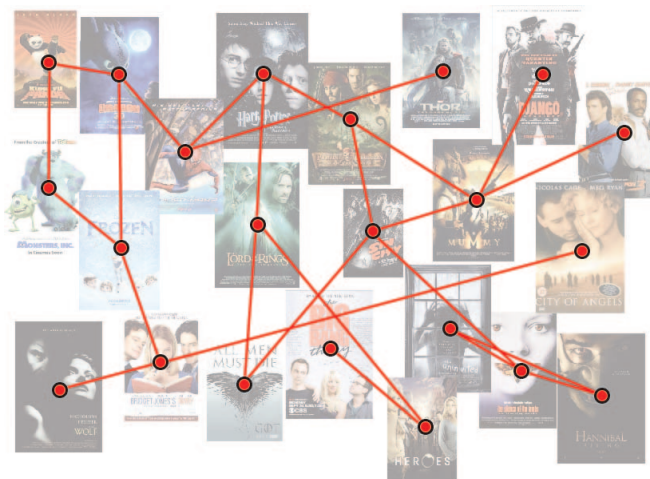


Graphs from **similarity** networks

[illegible]

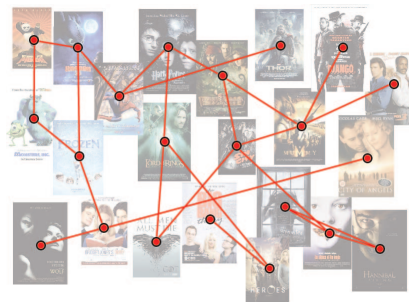
Graphs from **similarity networks**

and use it as an abstraction



Graphs from **similarity networks**

- ▶ vision
- ▶ audio
- ▶ text
- ▶ typical ML tasks
 - ▶ semi-supervised learning
 - ▶ spectral clustering



Movie similarity

Two sources of graphs in ML

Graph as models for networks

- ▶ given as an input
- ▶ discover interesting properties of the structure
- ▶ represent useful information (viral marketing)
- ▶ be the object of study (anomaly detection)

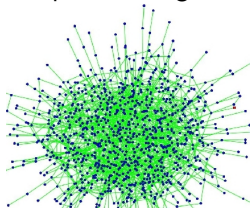
Graph as nonparametric basis

- ▶ we create (learn) the structure
- ▶ flat vectorial data $\rightarrow$ similarity graph
- ▶ nonparametric regularizer
- ▶ encode structural properties: smoothness, independence, ...

Random Graph Models

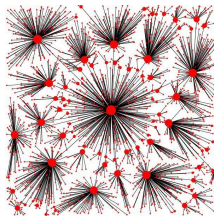
Erdős-Rényi

independent edges



Barabási-Albert

preferential attachment



Stochastic Blocks

modeling communities



Watts-Strogatz, Chung-Lu, Fiedler,

What will you learn in the Graphs in ML course?

Concepts, tools, and methods to work with graphs in ML.

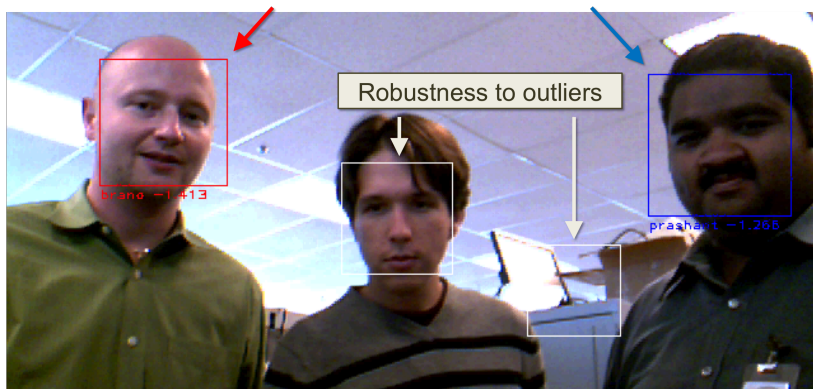
Theoretical toolbox to analyze graph based algorithms.

Specific applications of graphs in ML.

One example: Online Semi-Supervised Face Recognition

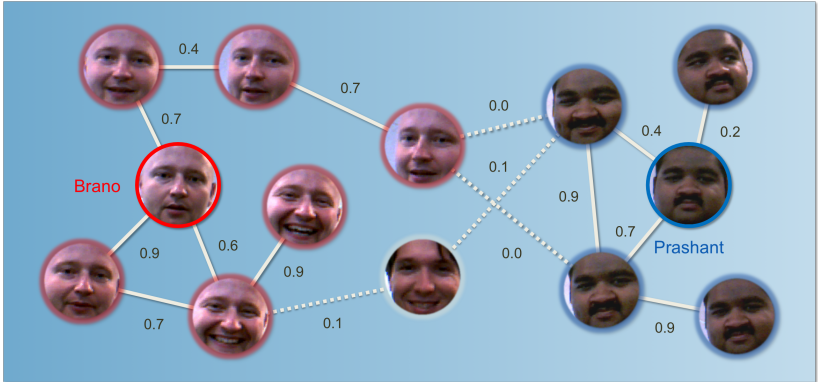
Online Semi-Supervised Face Recognition

graph is not given



Online Semi-Supervised Face Recognition

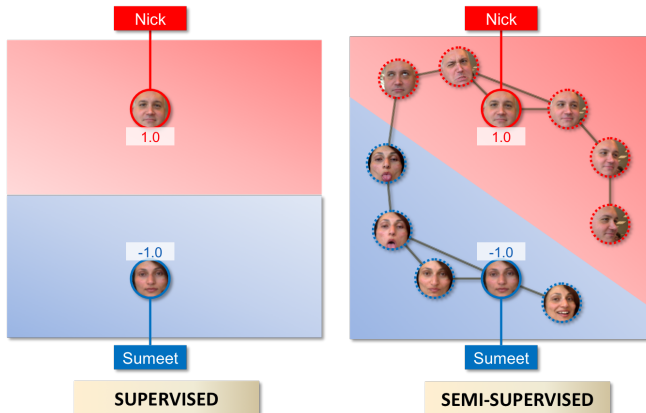
we will construct it!



An example of a similarity graph over faces. The faces are vertices of the graph. The edges of the graph connect similar faces. Labeled faces are outlined by thick solid lines.

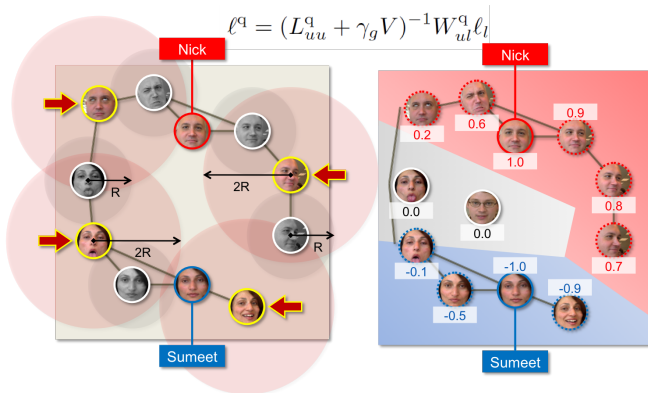
Online Semi-Supervised Face Recognition

graph-based semi-supervised learning



Online Semi-Supervised Face Recognition

online learning - graph sparsification



DEMO

second TD

see the demo: <http://researchers.lille.inria.fr/~valko/hp/serve.php?what=publications/kveton2009nipsdemo.officespace.mov>

OSS FaceReco: Analysis

$$\frac{1}{n} \sum_t (\ell_t^q[t] - y_t)^2 \leq \frac{3}{n} \sum_t (\ell_t^* - y_t)^2 + \frac{3}{n} \sum_t (\ell_t^o[t] - \ell_t^*)^2 + \frac{3}{n} \sum_t (\ell_t^q[t] - \ell_t^o[t])^2$$

Error of our
solution

Offline
learning error

Online learning
error

Quantization error

Claim: When the regularization parameter is set as $\gamma_g = \Omega(n_l^{3/2})$, the difference between the risks on labeled and all vertices decreases at the rate of $O(n_l^{-1/2})$ (with a high probability)

$$\frac{1}{n} \sum_t (\ell_t^* - y_t)^2 \leq \frac{1}{n_l} \sum_{i \in \mathcal{I}} (\ell_i^* - y_i)^2 + \beta + \sqrt{\frac{2 \ln(2/\delta)}{n_l}} (n_l \beta + 4)$$

$$\beta \leq \left[\frac{\sqrt{2}}{\gamma_g + 1} + \sqrt{2n_l} \frac{1 - \sqrt{c_u}}{\sqrt{c_u}} \frac{\lambda_M(L) + \gamma_g}{\gamma_g^2 + 1} \right]$$

OSS FaceReco: Analysis

$$\frac{1}{n} \sum_t (\ell_t^q[t] - y_t)^2 \leq \frac{3}{n} \sum_t (\ell_t^* - y_t)^2 + \frac{3}{n} \sum_t (\ell_t^o[t] - \ell_t^*)^2 + \frac{3}{n} \sum_t (\ell_t^q[t] - \ell_t^o[t])^2$$

Error of our
solution

Offline
learning error

Online learning
error

Quantization error

Claim: When the regularization parameter is set as $\gamma_g = \Omega(n^{1/4})$, the average error between the offline and online HFS predictions decreases at the rate of $O(n^{-1/2})$

$$\frac{1}{n} \sum_t (\ell_t^o[t] - \ell_t^*)^2 \leq \frac{1}{n} \sum_t \|\ell_t^o[t] - \ell^*\|_2^2 \leq \frac{4n_l}{(\gamma_g + 1)^2}$$

$$\|\ell\|_2 \leq \frac{\|y\|_2}{\lambda_m(C^{-1}K + I)} = \frac{\|y\|_2}{\lambda_m(K)\lambda_M^{-1}(C) + 1} \leq \frac{\sqrt{n_l}}{\gamma_g + 1}$$

OSS FaceReco: Analysis

$$\frac{1}{n} \sum_t (\ell_t^q[t] - y_t)^2 \leq \frac{3}{n} \sum_t (\ell_t^* - y_t)^2 + \frac{3}{n} \sum_t (\ell_t^o[t] - \ell_t^*)^2 + \frac{3}{n} \sum_t (\ell_t^q[t] - \ell_t^o[t])^2$$

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Quantization error

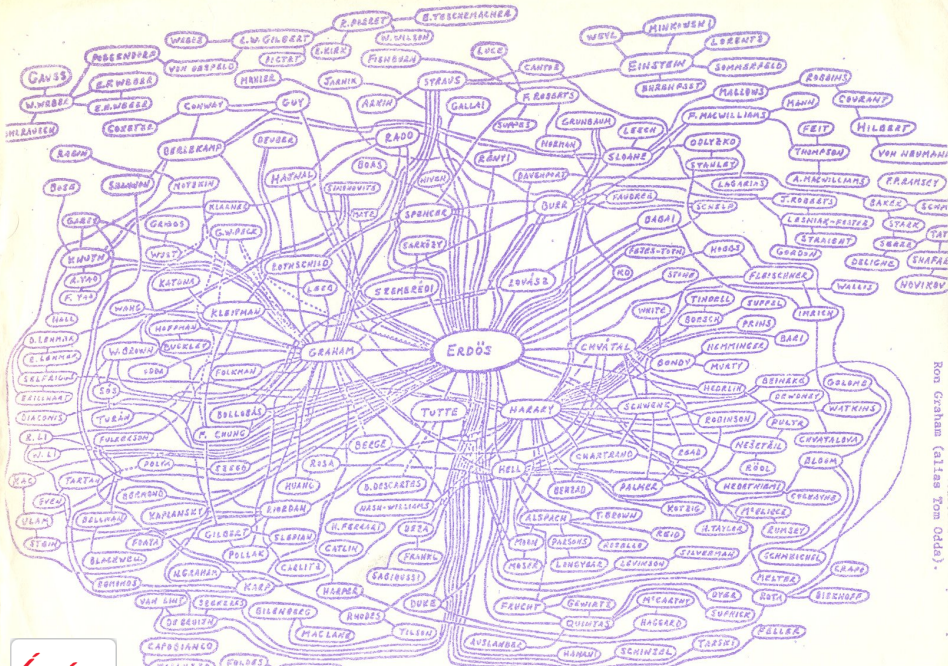
Claim: When the regularization parameter is set as $\gamma_g = \Omega(n^{1/8})$, and the Laplacians L^q and L^o are normalized, the average error between the online and online quantized HFS predictions decreases at the rate of $O(n^{-1/2})$

$$\frac{1}{n} \sum_t (\ell_t^q[t] - \ell_t^o[t])^2 \leq \frac{1}{n} \sum_t \|\ell_t^q[t] - \ell_t^o[t]\|_2^2 \leq \frac{n_t}{c_u^2 \gamma_g^4} \|L^q - L^o\|_F^2$$

$$\|L^q - L^o\|_F^2 \propto O(k^{-2/d})$$

The distortion rate of online k-center clustering is $O(k^{-1/d})$, where d is dimension of the manifold and k is the number of representative vertices

Erdős number and collaboration network



Ron Graham (alias Tom Odds).

Erdős number project

- ▶ <http://www.oakland.edu/enp/>
- ▶ example of a real-world graph
- ▶ 401 000 authors, 676 000 edges ($\ll 401000^2 \rightarrow$ sparse)
- ▶ average degree 3.36
- ▶ average distance for the largest component: 7.64
- ▶ 6 degrees of separation [Travers & Milgram, 1967]
- ▶ heavy tail

Some of the other topics

- ▶ link prediction/link classification
- ▶ signed networks (eOpinions)
- ▶ online decision-making on graphs
- ▶ submodularity on graphs
- ▶ real world graphs scalability and approximations
- ▶ spectral sparsification
- ▶ recommender systems applications
- ▶ large graph analysis, learning, and mining
- ▶ generalization bounds by perturbation analysis

Time: Tuesdays 11h-13h

Place: ENS Cachan - Room C103 (Bldg. Cournot, 1. floor)

8 lectures: 13.1. 20.1. 27.1. 10.2. 17.2. 3.3. 17.3. 24.3.

3 recitations (TD): 3.2. 24.2. 10.3.

Validation: grades from TD + class project

Course website:

<http://researchers.lille.inria.fr/~valko/hp/mva-ml-graphs>

Contact:

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TA: Daniele.Calandriello @ inria.fr

Sequel – INRIA Lille

MVA 2014/2015

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